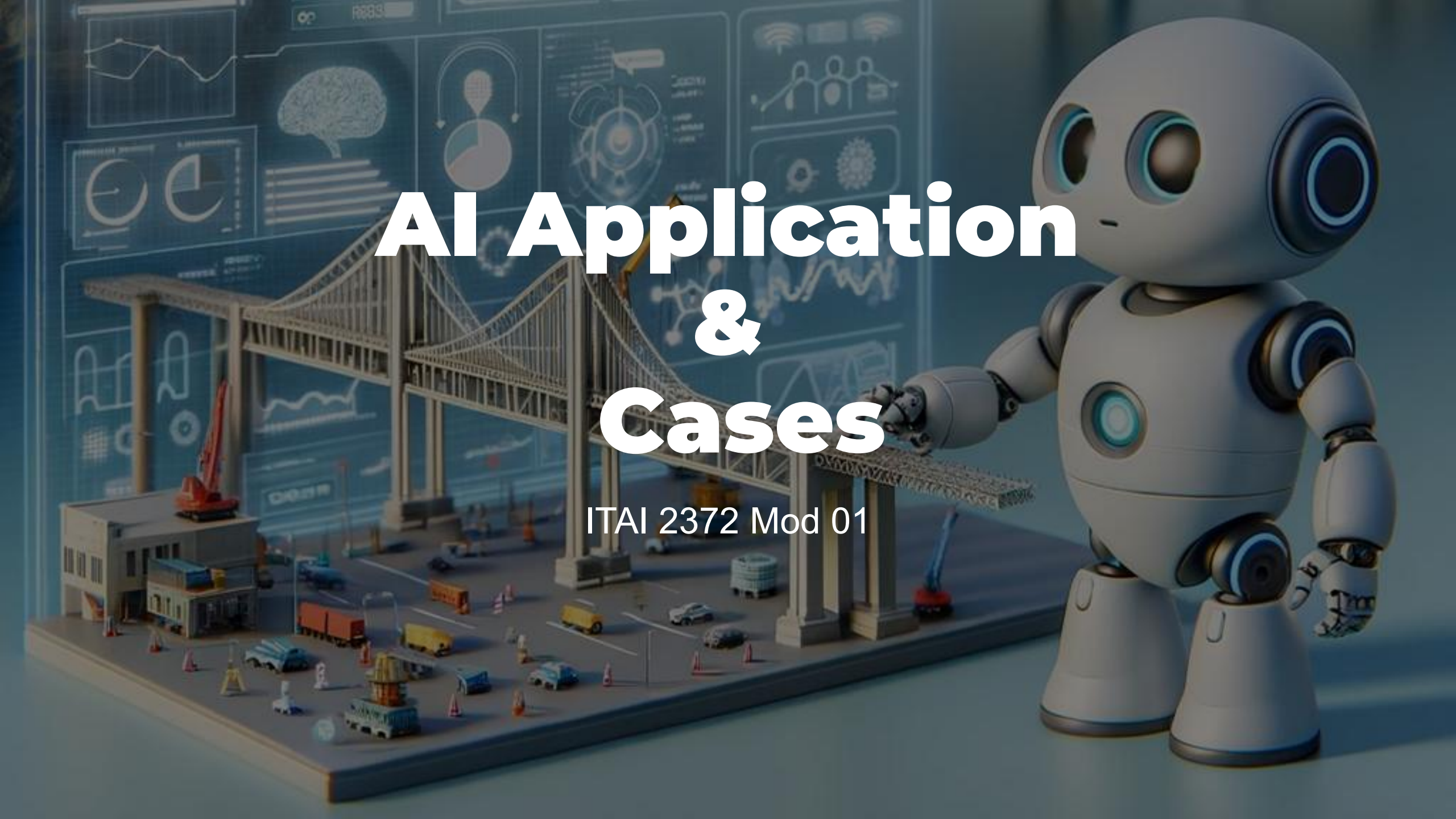


AI Application & Cases

ITAI 2372 Mod 01

Agenda

Information about the course

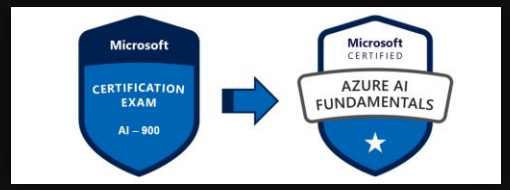
Todays (Mod 01) Topics

- The Role of Philosophy in Understanding AI
- Philosophical Questions about Consciousness and Intelligence
- Ethical Frameworks for AI Development
- Bias and Fairness in AI Algorithms
- The Impact of AI on Human Identity and Society
- The Future of Work and AI's Impact on Employment
- AI Governance and Policy Considerations

Welcome

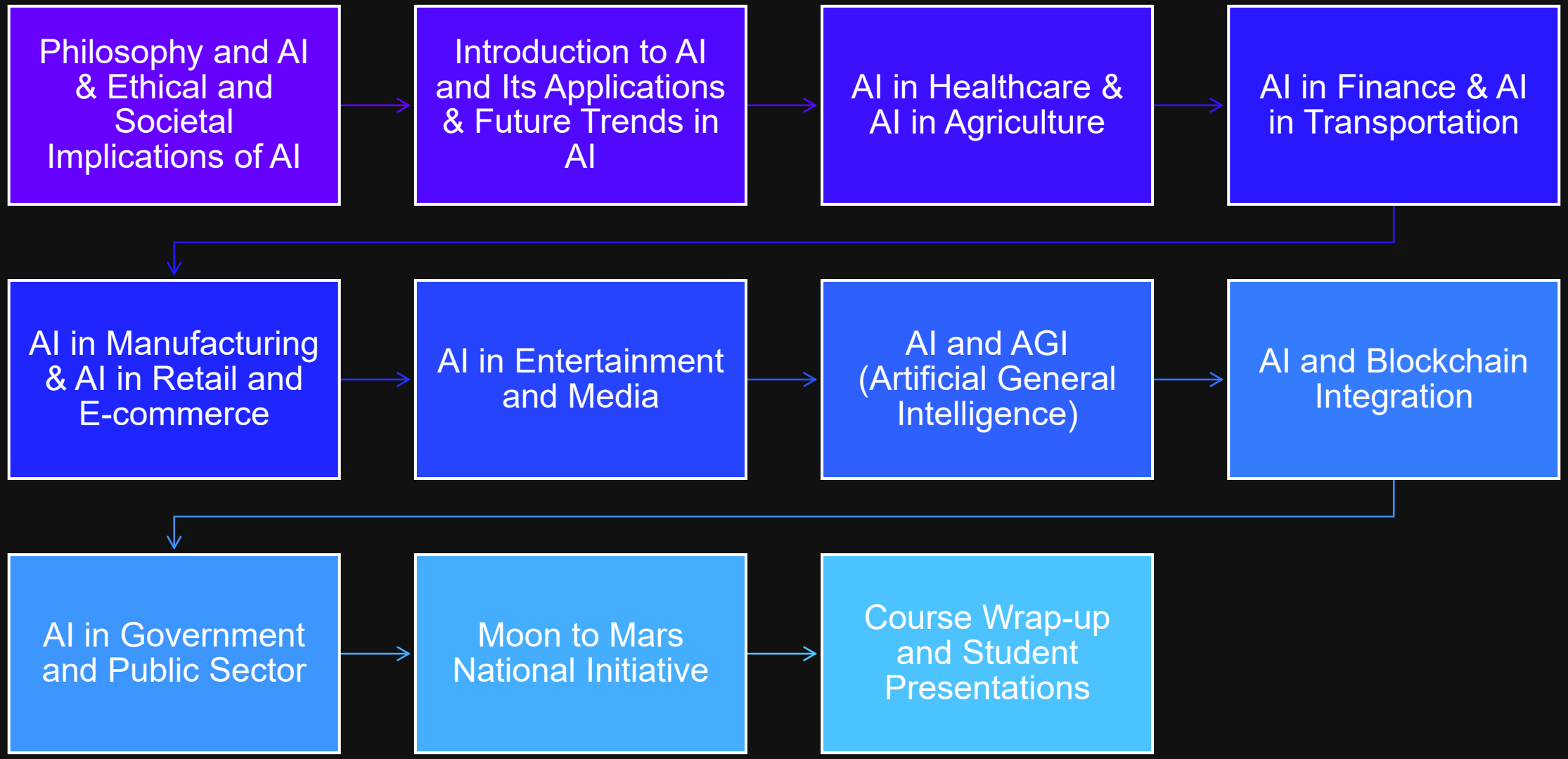
- What You'll Explore:
  - Real-world AI applications across industries like healthcare, finance, and entertainment.
  - Ethical, societal, and philosophical aspects of AI.
  - Case studies and group projects to enhance your skills in analyzing AI technologies.
- Important Note: This course adds the Microsoft AI-900 Certification! You'll have access to training slides and labs to prepare for the AI-900 exam and a 50% off voucher for the exam.

Course Calendar and Microsoft AI-900 Module



- **Weekly Topics:**
  - Each week will cover AI applications in various industries, with detailed case studies and discussions.
  - Familiar topics: Many applications were covered in other courses within the AI degree program.
- **Microsoft AI-900 Certification:**
  - A dedicated module on AI Fundamentals aligned with the AI-900 certification.
  - Access to Microsoft's AI-900 official training slides.
  - You will also get hands-on experience through AI-900 certification labs to solidify your knowledge.

Course Content Overview



Assignments Breakdown

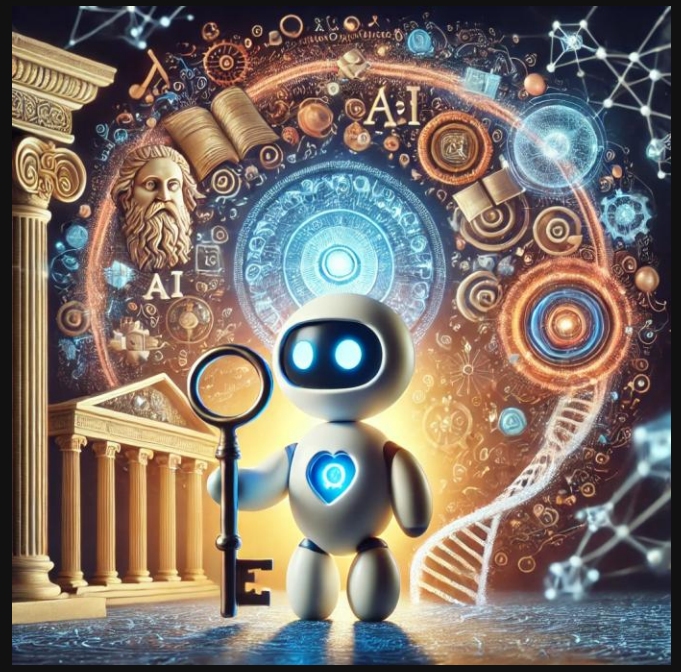
- Group Assignments (15%): Written Word documents, PDFs, or multimedia submissions
- Case Study Analysis (20%): Analysis based on lecture topics
- Exams/Quizzes (20%): Multiple-choice, true/false, or short-answer questions
- Midterm Project (20%): Group case study analysis
- Final Project (25%): Group proposal for integrating AI into a chosen sector
- Extra Credit Portfolio (5%): Uploading course portfolio to GitHub

The Role of Philosophy in Understanding AI (1 of 2)

- Philosophy as the foundation for exploring AI concepts.
  - **Ontology**: What is the nature of existence? How does AI fit in?
  - **Epistemology**: What can we know, and how do we know it? The role of data and reasoning in AI.
  - **Ethics**: What ought we to do? The moral implications of AI.

The Role of Philosophy in Understanding AI (2 of 2)

- Key Philosophical Movements:
  - Rationalism: The role of logic and reasoning in AI.
  - Empiricism: The importance of data and experience in shaping AI.
  - Existentialism: The search for meaning in AI and its applications.
- Philosophical Theories Applied to AI:
  - Dualism: AI as a non-conscious agent.
  - Materialism: AI as a product of physical processes.
  - Functionalism: AI as a system performing tasks based on its structure.



Philosophical Questions about Consciousness and Intelligence (1 of 2)

What is Consciousness?

- Consciousness as self-awareness and subjective experience.
- Can AI experience Qualia? The debate over subjective experience in machines.

The Turing Test:

- Measuring AI's ability to exhibit intelligent behavior.
- The difference between simulating intelligence and possessing it.

Philosophical Questions about Consciousness and Intelligence (2 of 2)

- **Searle's Chinese Room Argument**:
  - Understanding vs. Simulating: Can AI truly understand?
  - The limits of symbolic manipulation in AI.
- **Theories of Mind**:
  - **Dualism vs. Physicalism**: Can consciousness be separated from the physical?
  - **The Hard Problem of Consciousness**: Why does consciousness arise from physical processes?
  - **Integrated Information Theory (IIT)**: A framework for understanding consciousness in systems.

Ethical Frameworks for AI Development

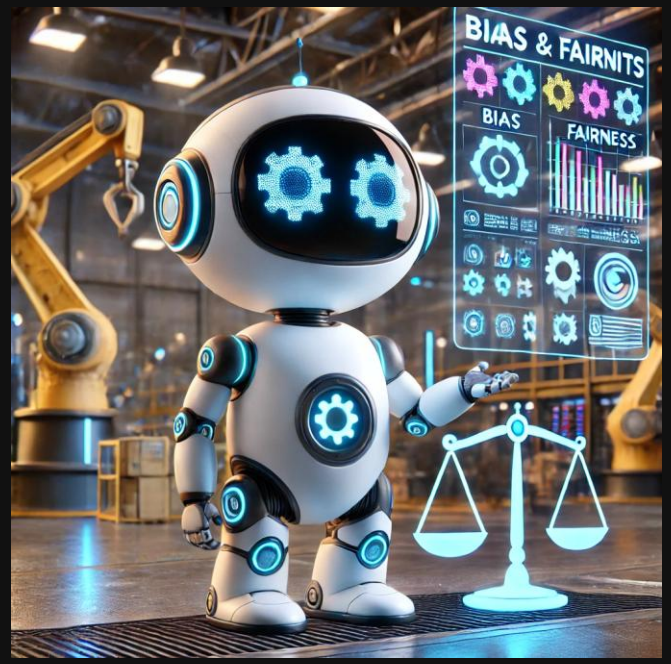
Utilitarianism: Maximizing happiness and reducing harm.

Deontology: Following moral rules regardless of outcomes.

Virtue Ethics: Focusing on the moral character of AI systems.

Case Study: Autonomous vehicles and ethical dilemmas.

Bias and Fairness in AI Algorithms



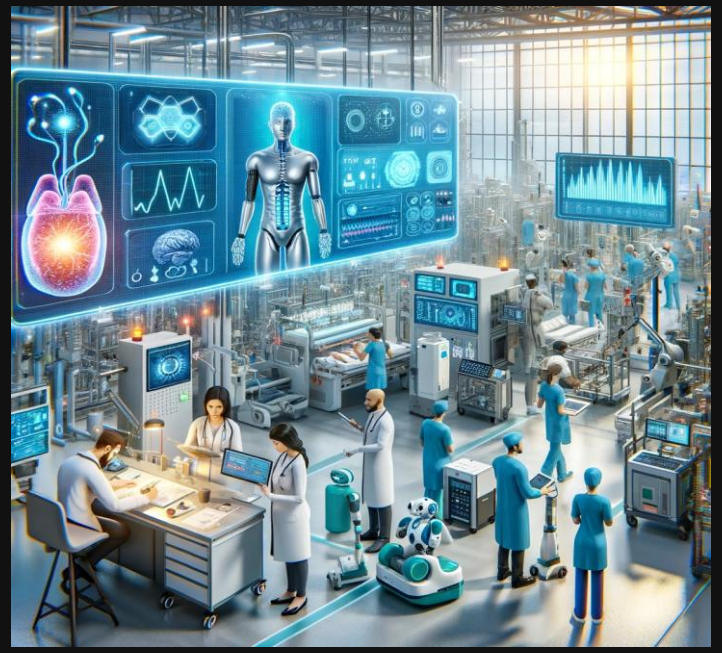
- **Sources of Bias**:
  - Historical data and training models.
  - Developer assumptions and design choices.
- **Fairness Measures**:
  - Demographic parity, equal opportunity, and calibration.
- **Impact of Biased AI**:
  - Case studies in hiring, facial recognition, and criminal justice.

The Impact of AI on Human Identity and Society

- AI and Privacy:
  - The erosion of privacy in the digital age.
- Shifts in Human Identity:
  - Digital twins, online personas, and AI companions.
- AI's Role in Reshaping Social Norms:
  - Influence on public discourse and cultural values.

The Future of Work and AI's Impact on Employment

- Automation and Job Displacement:
  - The potential for AI to replace certain types of jobs.
- The Rise of New Industries:
  - AI-enhanced roles and opportunities.
- Policies for a Fair Transition:
  - Universal basic income, retraining programs, and job guarantees.



AI Governance and Policy Considerations

- International AI Governance:
  - Balancing innovation with regulation.
- The Role of Public Policy:
  - Ensuring ethical AI development and deployment.
- Case Study: GDPR and its implications for AI.

Key Takeaways

- Philosophy and AI:
  - Philosophy provides the tools to explore fundamental questions about AI, such as its nature, knowledge, and ethical implications.
- Consciousness and Intelligence:
  - AI raises profound questions about consciousness and intelligence, challenging us to rethink what it means to be ‘aware’ and ‘intelligent.’
- Ethical and Societal Impacts:
  - The development and deployment of AI must be guided by ethical considerations, fairness, and an understanding of its impact on society.
- Governance and Policy:
  - Effective governance and international cooperation are essential to ensuring that AI benefits humanity while minimizing risks.